

Pricing AI as a Service

ILAN LOBEL¹, YANWEI SUN², JIAHUA WU²

NYU Stern School of Business¹, Imperial Business School²

I. MOTIVATION

AI pricing is moving beyond tokens!

- Foundation-model APIs mostly charge for usage, e.g., input/output tokens.
- Application-layer AI companies now charge for outcomes, such as resolved customer-service conversations.

Two pricing mechanisms.

Pay per usage (conversation-based pricing)
count tokens, API calls, conversations;
bill follows service use
examples: Salesforce Agentforce; Copilot Studio

VS.

Pay per success (outcome-based pricing) 📌
count reported resolutions;
bill follows reported success
examples: Fin, Zendesk, Sierra

What counts as success?

- Fin/Intercom counts no further human-help request after the last AI answer; Zendesk emphasizes “no human intervention.”
- Trust problem: Sierra notes concerns over “criteria to confirm an outcome.”

Two research questions.

- What contract and AI signaling mechanism maximize revenue?
- When does CBP beat OBP, or vice versa?

II. MODEL SETUP

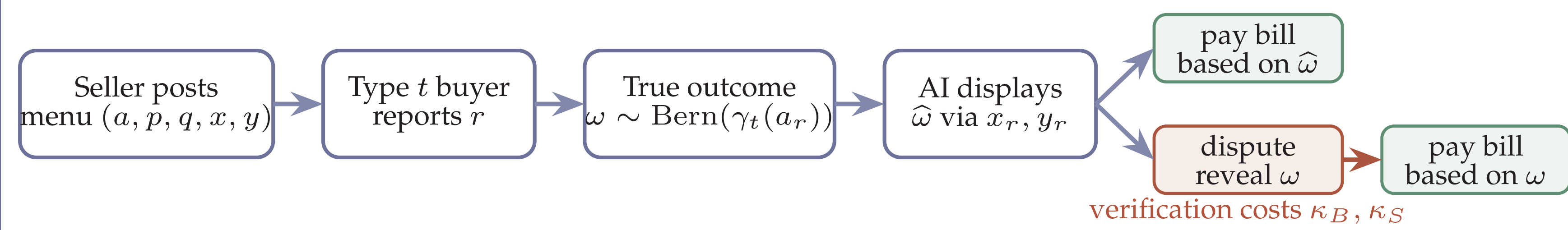
- Seller** offers AI models/agents with capability $a \in \mathcal{A}$ at cost $c(a)$ (increasing, convex).
- Buyer** has private type $t \in \mathcal{T} = \{1, \dots, n\}$, prior $\pi_t > 0$.
- Binary outcome:** $\omega \in \{\omega_0, \omega_1\}$ (failure/success).
- Type t buyer:** expected value $v_t(a)$, success probability of the outcome $\gamma_t(a)$.
 - We allow general $v_t(\cdot)$ and $\gamma_t(\cdot)$ functions (see V for a queueing application).
 - A *static* motivation: $v_t(a) := \gamma_t(a) \tilde{v}_t(a, \omega_1) + (1 - \gamma_t(a)) \tilde{v}_t(a, \omega_0)$.

Seller chooses two objects for each reported type t :

- Contract:** (a_t, p_t, q_t) . Capability a_t ; bills p_t, q_t after success/failure.
- AI signaling mechanism:** (x_t, y_t) , ex ante fine-tuning the AI model to potentially misreport:

$$x_t = \Pr(\hat{\omega} = \omega_1 \mid \omega = \omega_1, t), \quad y_t = \Pr(\hat{\omega} = \omega_1 \mid \omega = \omega_0, t).$$

x_t : truthful success display; y_t : false-success display, i.e. failure reported as success.



Seller's problem.

$$\max_{\{(a_t, p_t, q_t, x_t, y_t)\}_{t \in \mathcal{T}}} \sum_t \pi_t \left[\mathbb{E}\{\text{final bill paid by type } t\} - \kappa_S \mathbb{E}\{\text{dispute prob.}\} - c(a_t) \right].$$

- s.t. [IC] truthful report beats any misreport;
[SR] after observing $\hat{\omega}$, buyer optimally chooses pay directly or dispute;
[IR] expected utility ≥ 0 .

Truthful revealing benchmark ($x_t = 1, y_t = 0$).

$$\begin{aligned} & \max_{\{(a_t, p_t, q_t)\}_{t \in \mathcal{T}}} \sum_t \pi_t \left[(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t - c(a_t) \right] \\ \text{s.t. [IC]} & \quad v_t(a_t) - [(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t] \\ & \quad \geq v_t(a_r) - [(1 - \gamma_t(a_r)) q_r + \gamma_t(a_r) p_r], \quad \forall t, r \in \mathcal{T}, \\ \text{[IR]} & \quad v_t(a_t) - [(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t] \geq 0, \quad \forall t \in \mathcal{T}. \end{aligned}$$

III. OPTIMAL MECHANISM

Separable primitives. $v_t(a) = \theta_t h(a)$, $\gamma_t(a) = \alpha_t s(a)$, with h increasing and concave.

Theorem (Optimal mechanism)

- Signaling mechanism:** truthful revelation is without loss.
- Contract:** under separable primitives (order types by $\alpha_1 < \dots < \alpha_n$).
 - Convex formulation** of the *bi-linear* program, based on the geometric interpretation of IC, with allocations and buyer surplus as decision variables.
 - Endpoint success-probability types get first-best allocations.** The lowest- and highest-success-probability types receive first-best allocations:

$$a_1^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{\theta_1 h(a) - c(a)\}, \quad a_n^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{\theta_n h(a) - c(a)\}.$$

- Curvature virtual types.** For $t = 2, \dots, n - 1$,

$$\hat{\theta}_t = \theta_t - \frac{1}{\pi_t} \sum_{r < t < \tau} \mu_t^{r\tau} \Delta_t^{r\tau}, \quad a_t^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{\hat{\theta}_t h(a) - c(a)\}.$$

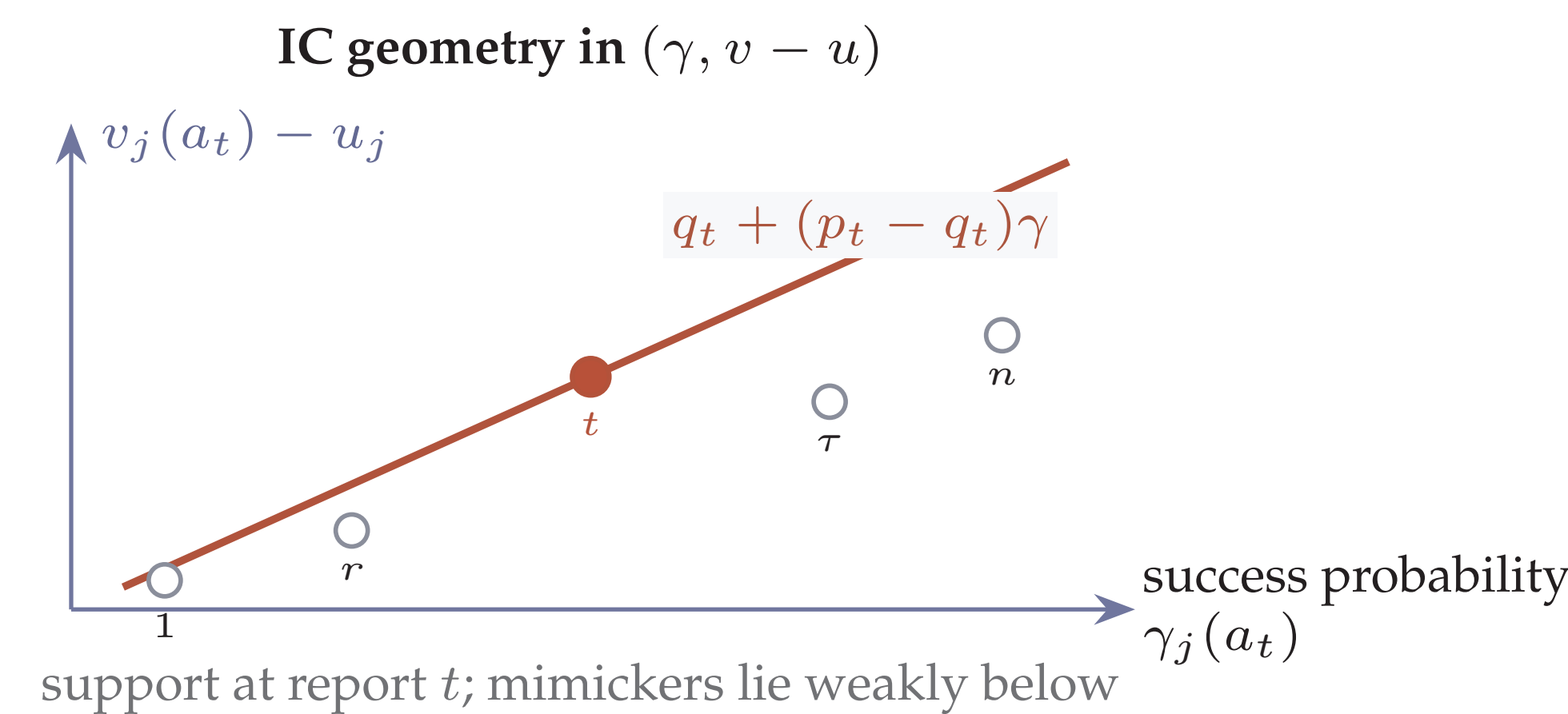
- * $\Delta_t^{r\tau}$: measures the curvature of the three points $(\alpha_r, \theta_r), (\alpha_t, \theta_t), (\alpha_\tau, \theta_\tau)$; $\Delta_t^{r\tau} < 0$ means concave, $\Delta_t^{r\tau} > 0$ means convex.
- * $\mu_t^{r\tau} \geq 0$: Lagrange multiplier on the IC constraint that prevents types $r < t$ and $\tau > t$ from mimicking report t .
- * Interior distortions go both ways: $a_t^{\text{SB}} < a_t^{\text{FB}}$ and $a_t^{\text{SB}} > a_t^{\text{FB}}$ can both happen.

- Truthful revelation:**
 - Signaling makes the buyer's cost *concave* in success probability.
 - Concavity implies the existence of a supporting line at the true type that preserves the buyer's cost and lies above all mimickers, making mimicry no cheaper.
- Geometry of IC constraints** (under truthful revelation):

Choose allocation-buyer-surplus pairs (a_t, u_t) :

$$\max_{\{a_t, u_t\}_{t \in \mathcal{T}}} \sum_t \pi_t [v_t(a_t) - c(a_t) - u_t].$$

[IC] buyer t chooses item (a_t, u_t) .
[IR] $u_t \geq 0$ for every t .



- Bills (p_t, q_t) create one expected-payment line in γ , $q_t + (p_t - q_t)\gamma$.
- Passing through report t preserves its assigned bill; staying above other points makes mimics no cheaper.
- Middle reports face mimicry from both sides: IC requires an upper-support geometry; see the figure.
- Endpoint reports face only one-sided mimicry: because there is no type beyond the boundary, a one-sided support line can pass through the endpoint and keep all possible mimics below.

Illustration of how type geometry determines the second-best allocation:

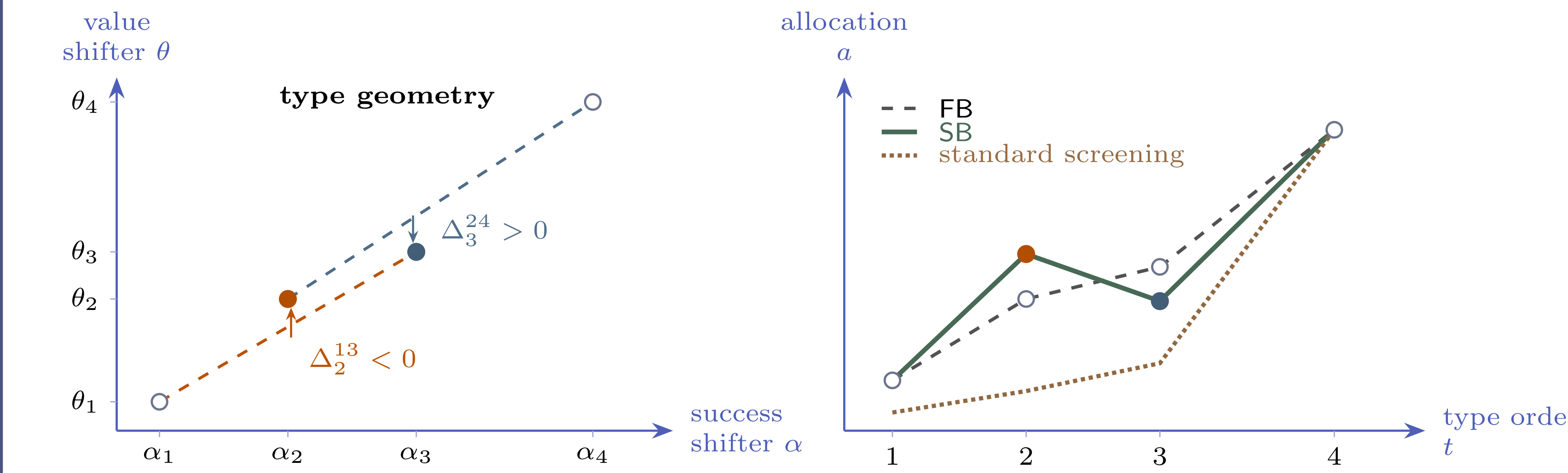


Figure note. Types are ordered so $\alpha \uparrow$ also means $\theta \uparrow$. Left: local curvature gaps for the ordered points (α_t, θ_t) . Right: FB rises with θ ; standard screening (CBP) also has $a \uparrow$ with θ but lies below FB for non-top types; unrestricted SB keeps endpoints first-best, while $\Delta < 0$ raises and $\Delta > 0$ lowers interior allocations.

Corollary (First-best extraction)

$R_{\text{SB}} = R_{\text{FB}}$ if and only if the first-best points satisfy the supporting-line conditions.

- Separable primitives: discrete concavity of (α_t, θ_t) suffices.
- Binary types: no two-sided IC, so first best is automatic.

IV. PRACTICAL DEBATE: CBP vs. OBP

Revenue difference decomposition.

$$R_{\text{OBP}}^* - R_{\text{CBP}}^* = \underbrace{R_{\text{OBP}}^{\text{tr}} - R_{\text{CBP}}^*}_{\text{baseline gap}} + \underbrace{R_{\text{OBP}}^* - R_{\text{OBP}}^{\text{tr}}}_{\text{manipulation gain} \geq 0}.$$

CBP: $p_t = q_t$
OBP: $q_t = 0$

- $R_{\text{OBP}}^{\text{tr}}$: OBP with truthful outcome revelation.

Theorem (OBP vs. CBP)

- Baseline gap:** $R_{\text{OBP}}^{\text{tr}} \geq R_{\text{CBP}}^*$ if per-success value $v_t(a)/\gamma_t(a)$ aligns with success probability.
- Manipulation gain:**

$R_{\text{OBP}}^* - R_{\text{OBP}}^{\text{tr}}$ is increasing in buyer-side dispute cost κ_B .

- Once the buyer-side dispute cost κ_B is high enough:

OBP (pay on displayed success) + signaling
 $\iff$ limited-liability outcome prices $(p, q \geq 0)$ + truthful revelation.

Proposition (Asymmetric worst-case robustness)

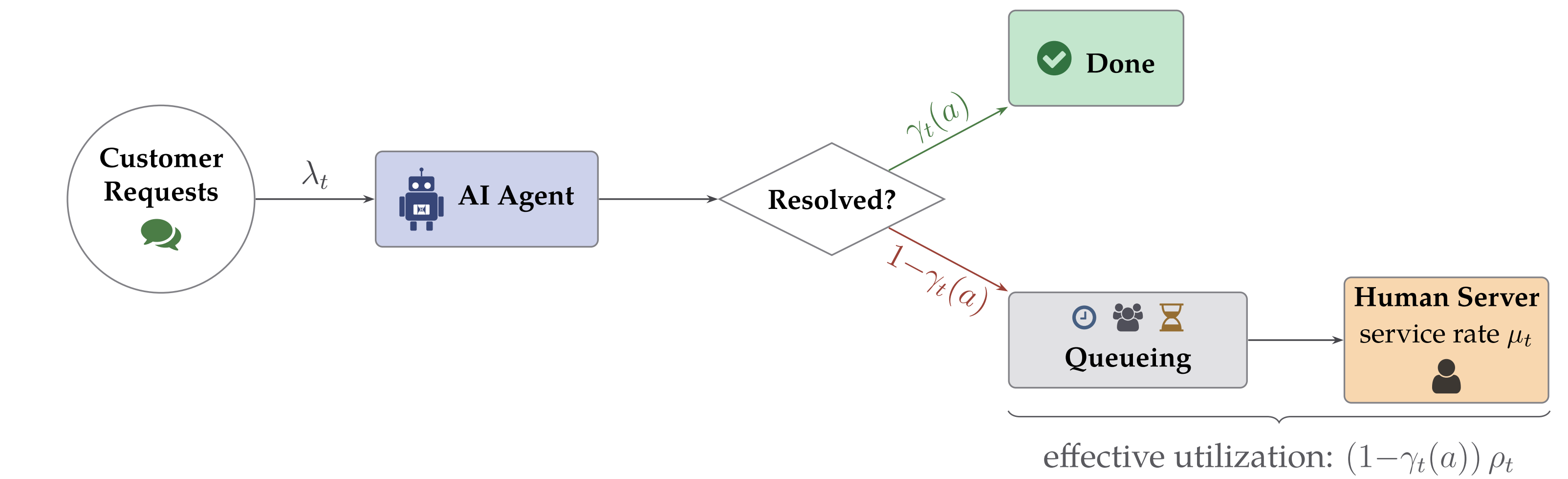
- CBP achieves at least $1/n$ of the second-best revenue:

$$R_{\text{CBP}}^* \geq \frac{1}{n} R_{\text{SB}}.$$

- OBP can be arbitrarily bad in the worst case ($R_{\text{CBP}}^*/R_{\text{OBP}}^*$ unbounded).

V. APPLICATION: PRICING AI IN CUSTOMER SERVICE

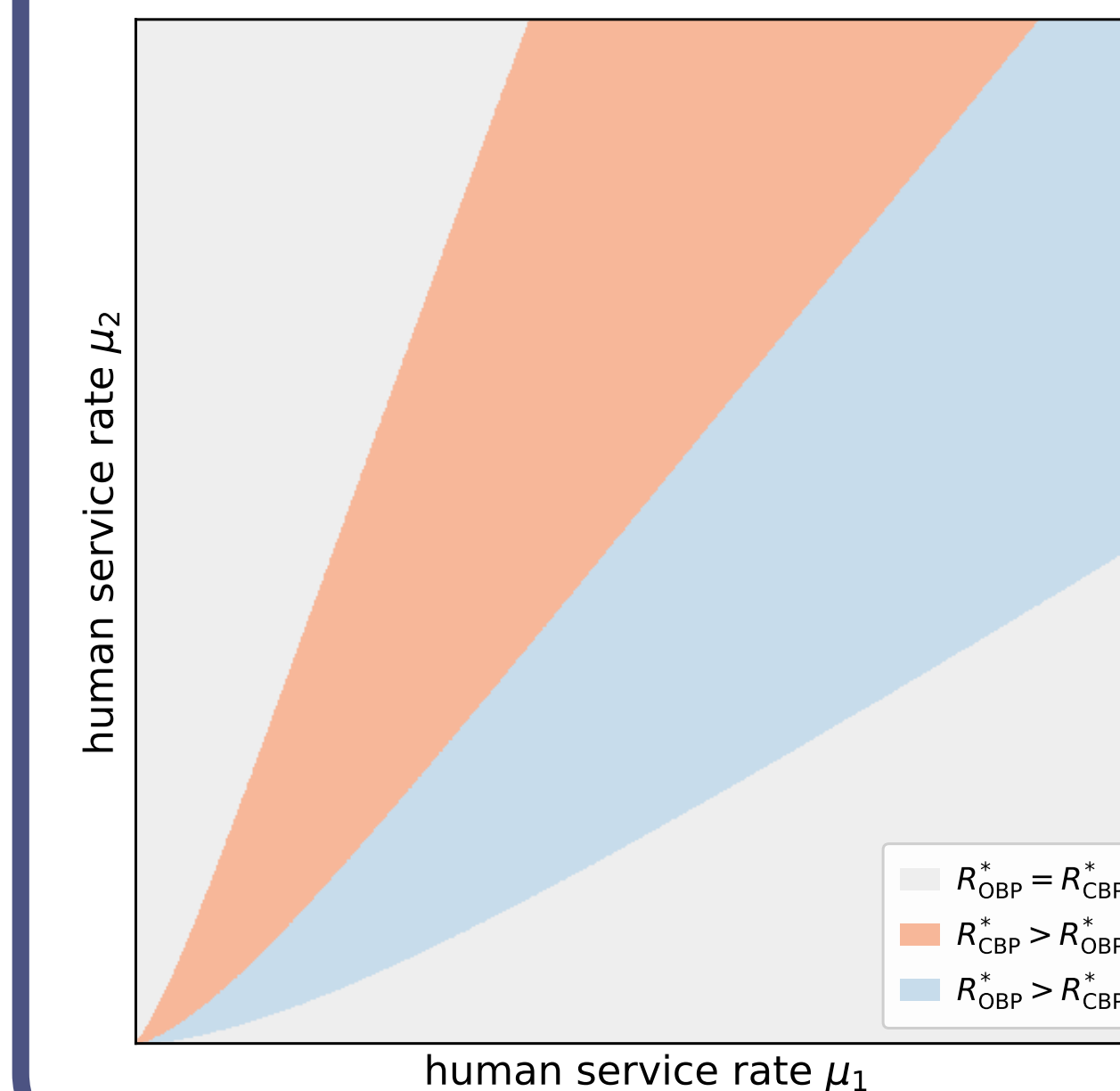
A firm (AI buyer) deploys an AI agent as a front end. Resolved requests exit; failures escalate to a human queue. (Assume truthful revelation here.)



- Private type determines $\lambda_t, \mu_t, \rho_t = \lambda_t/\mu_t$, and AI success $\gamma_t(a)$.
- Value $v_t(a)$ is congestion-cost reduction. For M/M/1 with holding cost h :

$$v(\rho_t, \gamma_t(a)) = h \left[\frac{\rho_t^2}{1 - \rho_t} - \frac{(1 - \gamma_t(a))^2 \rho_t^2}{1 - (1 - \gamma_t(a)) \rho_t} \right].$$

Revenue comparison illustration: binary types, $\gamma_2 > \gamma_1$, single allocation.



- & **Substitutes:** humans and AI excel at the same types; AI's edge is redundant $\Rightarrow$ **CBP** wins.
- & **Complements:** humans and AI excel at different types; each covers the other's weakness $\Rightarrow$ **OBP** wins.